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MAPLES-DR: MESSIDOR Anatomical and Pathological Labels for Explainable Screening of Diabetic Retinopathy

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Reliable automatic diagnosis of Diabetic Retinopathy (DR) and Macular Edema (ME) is an invaluable asset in improving the rate of monitored patients among at-risk populations and in enabling earlier treatments before the pathology progresses and threatens vision. However, the explainability of screening models is still an open question, and specifically designed datasets are required to support the research. We present MAPLES-DR (MESSIDOR Anatomical and Pathological Labels for Explainable Screening of Diabetic Retinopathy), which contains, for 198 images of the MESSIDOR public fundus dataset, new diagnoses for DR and ME as well as new pixel-wise segmentation maps for 10 anatomical and pathological biomarkers related to DR. This paper documents the design choices and the annotation procedure that produced MAPLES-DR, discusses the interobserver variability and the overall quality of the annotations, and provides guidelines on using the dataset in a machine learning context.

Background & Summary

Diabetic retinopathy (DR) is a complication of diabetes mellitus (DM) that damages the retinal microvasculature and can lead to vision impairment. DR develops gradually and is clinically characterized by stages according to the presence of lesions in the retina. Non-invasive fundus imaging can detect these lesions, and regular screening of at-risk populations is necessary to ensure early treatment and preserve their vision¹. Despite widespread screening programs in North America^{2–4}, 40% of patients with DM are still not monitored for DR. Studies based on these programs indicate that teleophthalmology screening increases the rate of monitored patients⁵, and that algorithms for automatic diagnosis of DR can enable further improvements by reducing costs and increasing the frequency of examinations, as well as ensuring timelier management of referred patients¹.

In the last decade, machine learning models have been successfully applied to the automatic diagnosis of DR in fundus images. At the core of the development of these supervised algorithms are public datasets of annotated images, used for training and validation⁶. Among them, the Eyepacs⁷ and MESSIDOR⁸ datasets have enabled the development of automatic screening algorithms that exceed the performance required by the FDA⁶, and even outperform human experts⁹. Yet, medical personnel still lack confidence in these technologies, which do not meet the standards of explainable AI. We believe that to overcome this limitation, improving training labels should not be overlooked. Indeed, uncertainty around validation methods and generalization issues call for more diverse and bias-aware validation datasets and better documentation of their labeling process. The weak explainability of screening models calls, in addition, for more exhaustive and clinically relevant training labels beyond simple diagnostic classification. In its report “Four principles of Explainable AI”, the National Institute of Standards and Technology underlines that a proper explanation of algorithm outcomes should be *meaningful* to the target audience¹⁰, namely, in our case, it must be formulated using a vocabulary familiar to ophthalmologists. Restricting training labels to only DR grades is a major shrinkage of medical vocabulary, especially when clinical

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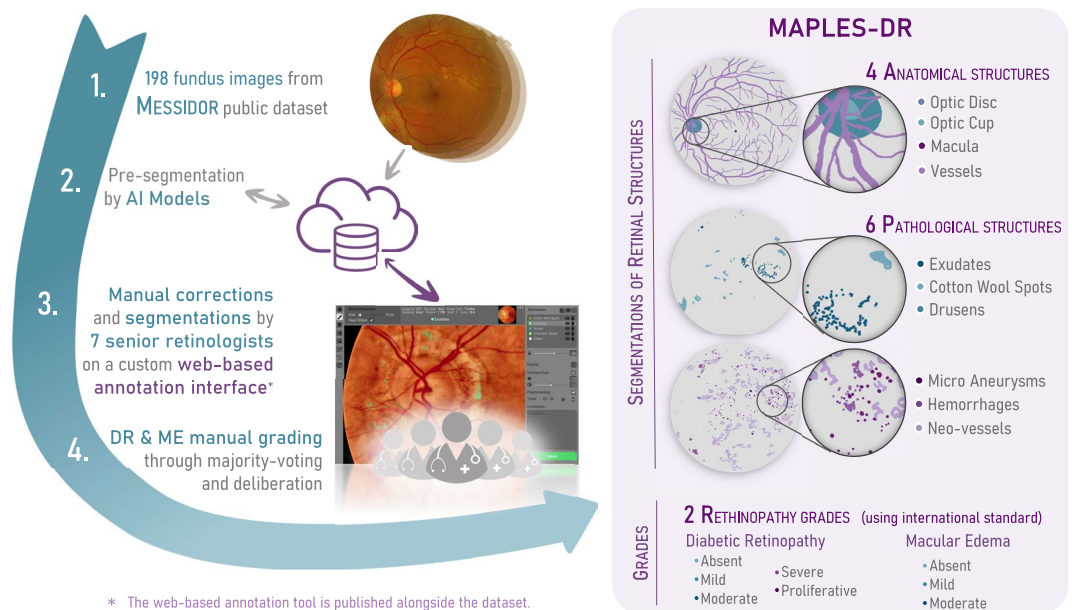


Fig. 1 Schematic overview of the MAPLES-DR annotation protocol and of the content of the dataset.

justifications rely heavily on identifying the anatomical and pathological structures of the retina (vessels, macula, red or bright lesions, etc.)^{11–13}. Therefore, whether it is to guide models *a priori* so that they learn representations compatible with clinical knowledge or to interpret these representations *a posteriori* by comparing them with known biomarkers, datasets with pixel-wise annotations of such structures play a key role in the development of explainable screening models for DR. We designed *MAPLES-DR* (MESSIDOR Anatomical and Pathological Labels for Explainable Screening of Diabetic Retinopathy) with this objective in mind.

Annotating pixel-wise labels requires considerably more time and effort than image-level diagnostic labels; consequently, fundus datasets with such labels are few. Most of them publish annotations for only one biomarker (e.g. vessels^{14–17}, optic disc^{18–20}, exudates²¹, microaneurysms²²). To our knowledge, only four datasets provide labels for four lesions that are symptomatic of DR: microaneurysms, hemorrhages, exudates, and cotton-wool spots (CWS). FGADR²³ and Retinal Lesions²⁴ comprise numerous images (1842 and 1593 respectively), but pathological structures are mostly annotated as large clusters of lesions. In contrast, IDRiD²⁵ and DDR²⁶ provide precise segmentations of each lesion, but for fewer images (81 and 757 images, respectively). The articles accompanying these datasets focus on the performance gain they bring to existing screening methods, but provide only shallow descriptions of the protocols used to produce their annotations. None of them contain annotations of anatomical structures.

The *MAPLES-DR* dataset extends the MESSIDOR⁸ dataset by complementing its image-wise diagnostic labels with novel pixel-level annotations of 10 classes of anatomical and pathological structures along with new DR and macular edema (ME) grades for 198 of its images (see Fig. 1). As far as we know, no other public dataset offers such a comprehensive range of anatomical and pathological labels for fundus images. This paper extensively documents the design choices and labeling protocol of this dataset, as well as its potential biases and inaccuracies. By thoroughly documenting our approach and releasing all the tools and software that produced the annotations along with this paper, we aim to help other research teams build more such datasets for explainable DR screening.

Methods

MAPLES-DR does not provide new fundus images, but extends MESSIDOR labels by releasing annotations of *new biomarkers* for 198 fundus images of the original MESSIDOR dataset⁸ published in 2014. The publication of this new annotations dataset was approved by the MESSIDOR consortium. This section describes the annotation process that produced *MAPLES-DR*.

Fundus images selection. Of the 1,200 images available in the original MESSIDOR dataset, 200 were selected for annotation in *MAPLES-DR*. We randomly chose 30 healthy images, 59 R1 images, 55 R2 images, and 56 R3 images, according to the DR grades provided by MESSIDOR. This selection is not representative of the prevalence of the pathology in either the MESSIDOR dataset or an actual screened population (see Fig. 2). However, it provides a set of images for each pathology stage and focuses on those critical for screening (R1 and R2). In the last stage of the annotation campaign, we found two duplicate entries in the dataset (the original MESSIDOR dataset contains several duplicate images stored with different names, which was later fixed in MESSIDOR-2²⁷). We excluded them from *MAPLES-DR*, reducing the number of images to 198.

Procedure to grade DR and ME. The diagnoses provided with the MESSIDOR database for DR and ME are based on an uncommon grading system. To facilitate comparison with other public fundus image databases,

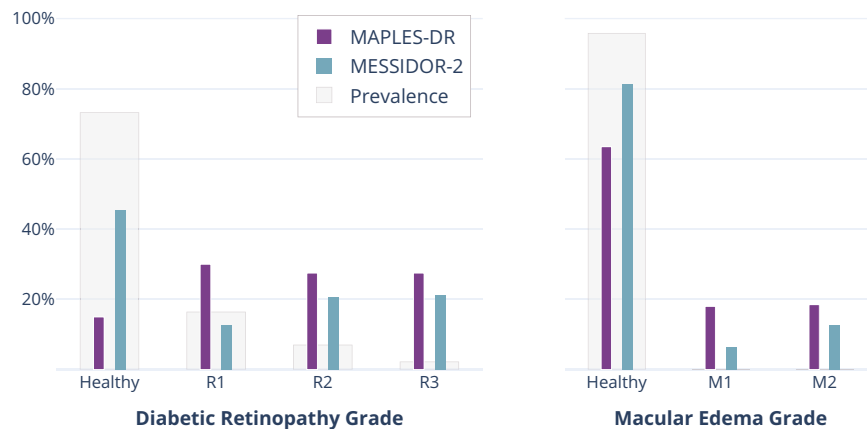


Fig. 2 Disparities among DR and ME grade distributions (converted to MESSIDOR grades) between MAPLES-DR subset (purple), MESSIDOR complete dataset (teal blue) and prevalence from first screening of the Toronto tele-ophthalmology program⁴⁰ (gray).

we regraded MAPLES-DR images following the guidelines for Canadian teleophthalmology screening, whose grades are closer to international standards (i.e. ICDR¹¹ or the Scottish system¹²).

These guidelines distinguish six grades for DR (R0: absent, R1: mild, R2: moderate, R3: severe, R4A: proliferative, R4S: stable treated proliferative) and three for ME (M0: absent, M1: mild, M2: moderate). Each grade is associated with a recommended course of action (from rescreening in 12–24 months to immediate referral to an ophthalmologist) and is determined by the number and position of red lesions (hemorrhages, microaneurysms, neovessels) and bright lesions (exudates, cotton wool spots) visible in the retina. Images with insufficient quality for diagnosis could also be flagged as R6 or M6. A comprehensive definition of this grading system was published by Boucher *et al.*¹³.

The variability between observers in the diagnosis of DR and ME is not negligible and is well documented in the literature^{28–32}. Our DR and ME grading procedures accounted for this by splitting diagnosis labeling into two phases. First, each image was individually graded by three senior retinologists and a common diagnosis was determined by majority voting. The few cases of complete disagreement between the three retinologists (5 cases of DR and 12 cases of ME out of 198 images) were resolved in a second phase by deliberation between the experts to establish a consensual diagnosis.

Procedure to annotate Retinal Structures. The main contribution of MAPLES-DR is the annotation of retinal structures symptomatic of DR at a pixel-level, which are absent from the original MESSIDOR dataset.

Selection of anatomical and pathological biomarkers. The decision to include or exclude retinal structures from the list of annotated biomarkers was based on clinical understanding of their role in DR histopathology and screening procedures. See Figs. 3 and 4 for examples of pathological structures as annotated in MAPLES-DR.

Diabetes mellitus affects the walls of the vessels, eventually causing microvascular dysfunctions that manifest in the retina as microaneurysms, hemorrhages, intraretinal microvascular abnormalities (IRMA), or neovessels. We refer to these pathological structures as “red lesions”. *Microaneurysms* appear as small circular dilations of the capillaries. They are early signs of microvascular dysfunction and are commonly used to detect mild DR. *Intraretinal Hemorrhages* develop in the more advanced stages of the pathology. They are usually divided into dot hemorrhages (circular and well-defined spots, typically caused by a microaneurysm rupture) and blot hemorrhages (larger with less defined borders). Both were simply annotated as hemorrhages in MAPLES-DR. *IRMA* can appear starting from the moderate non-proliferative stage (R2), but their proliferation coincides with the next stage of the disease (R3), indicating a 50% risk of developing neovascularization within one year. The emergence of *Neovessels* (NV) signals a transition to the proliferative stage (R4A), the most severe stage of the DR screening, which requires immediate referral to an ophthalmologist. Indeed, extensive NVs can leak, causing preretinal or vitreous hemorrhages and leading to major visual loss if left untreated. The distinction between NVs and IRMAs is difficult to establish using fundus images alone and normally requires fluorescein angiography. In the absence of this imaging modality, IRMAs are not differentiated from NVs in MAPLES-DR.

In the severe stages of DR, the retina thickens and “bright lesions” can appear. Hard *exudates* usually arise from leakage from damaged capillaries, potentially causing loss of visual acuity. Furthermore, ischemia may cause a blockage in axonal transport in the optic nerve fiber layer, which can in turn lead to axoplasmic accumulation and the appearance of lesions known as *Cotton Wool Spots* (CWS). These lesions are characterized by their white appearance and blurry borders. While the principal etiology is diabetic retinopathy, CWS can also be observed in other vascular diseases (systemic arterial hypertension, vein obstruction, coagulopathies, etc.). Finally, we provide annotations of *Drusens*. While these bright lesions are not symptomatic of DR and are more commonly associated with Age-related Macular Degeneration (AMD), their aspect is similar to exudates and can be mistaken for them. They usually appear around the macula and are histologically situated at the interface with the Retinal Pigment Epithelium (RPE). It is supposed that they originate from degenerative products of the RPE’s cells and are composed of lipids and glycoproteins.

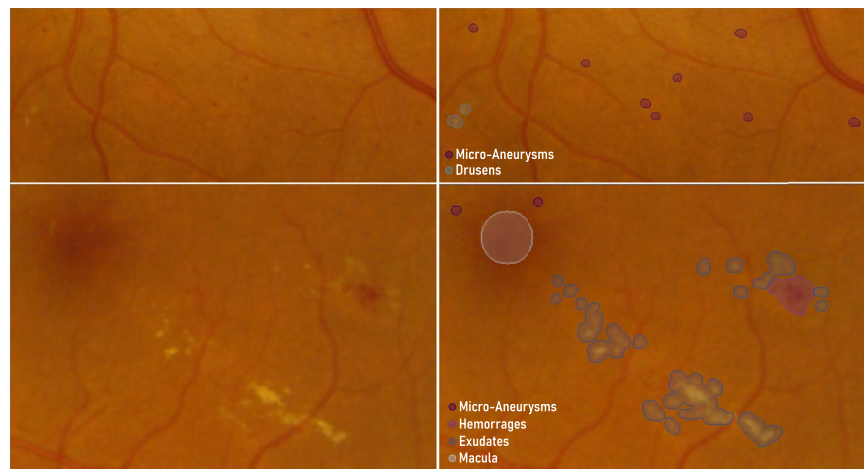


Fig. 3 Pathological structures from image 20060411_59190_0200_PP. (Left: the fundus image without annotations).

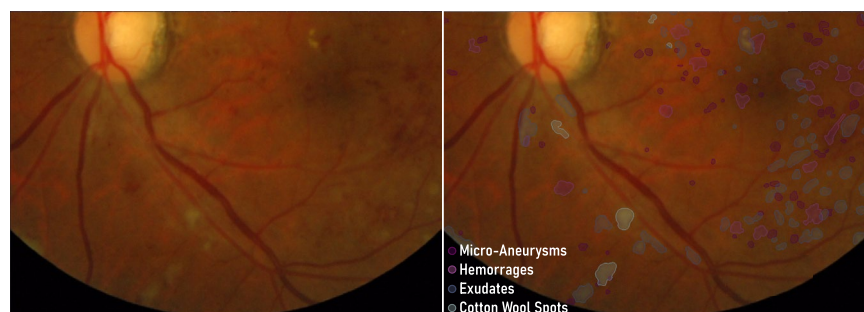


Fig. 4 Pathological structures from image 20051214_57940_0100_PP.

Besides pathological biomarkers, MAPLES-DR also includes segmentation of anatomical structures: the optic disc and cup, the macula, and retinal vessels. Although these anatomical structures are present in all images, including healthy ones, their appearance and proximity to lesions provide valuable diagnostic information. *Retinal vessels* morphology is indicative of stages of DR: an increase in arteriolar tortuosity is associated with mild and moderate stages³³, while venous beading and dilation are symptoms of severe proliferative stages. In addition, the severity of a lesion often depends on its position relative to the *Optic disc* or the *Macula*. ME is graded by counting the number of lesions within one or two optic disk diameters from the macula. Some clinical guidelines distinguish disc neovascularization from other neovascularization, etc. Fig. 5 shows an example of MAPLES-DR annotations of the retinal anatomical structures.

The choice of jointly annotating anatomical biomarkers and pathological biomarkers is, to our knowledge, unprecedented for fundus datasets, and distinguishes MAPLES-DR from existing datasets of DR annotations such as FGADR²³ or IDRiD²⁵.

Recruiting annotators and designing the annotation tools. The segmentation maps of MAPLES-DR were annotated by seven Canadian retinologists affiliated with five different hospitals: Hôpital Maisonneuve Rosemont, Montréal; University Health Network, Toronto; Centre Hospitalier Universitaire de Montréal; Université de Sherbrooke; and Centre Hospitalier Universitaire Saint-Justine, Montréal). All the annotators were seniors retinologists and were recruited thanks to their involvement in teleophthalmology programs for the detection of DR in Quebec and Ontario.

The annotations were collected on a custom web-based annotation platform that we developed specifically for this project. It was designed to support a “scalable” annotation protocol, capable of being extended to much more ambitious annotation campaigns, such as labeling large Canadian telemedicine databases containing tens of thousands of images. With this in mind, its web interface was designed to allow users to work collaboratively on the same images regardless of their geographical location. Its specialized drawing and visualization tools provided an intuitive yet effective workflow to segment biomarkers in fundus images.

Preprocessing MESSIDOR fundus images. The dimensions of MESSIDOR fundus images varies from 1440×960 to 2304×1536 pixels. To standardize the resolution of the MAPLES-DR segmentation maps, all fundus images were cropped and resized to 1500×1500 pixels prior to the annotation process.

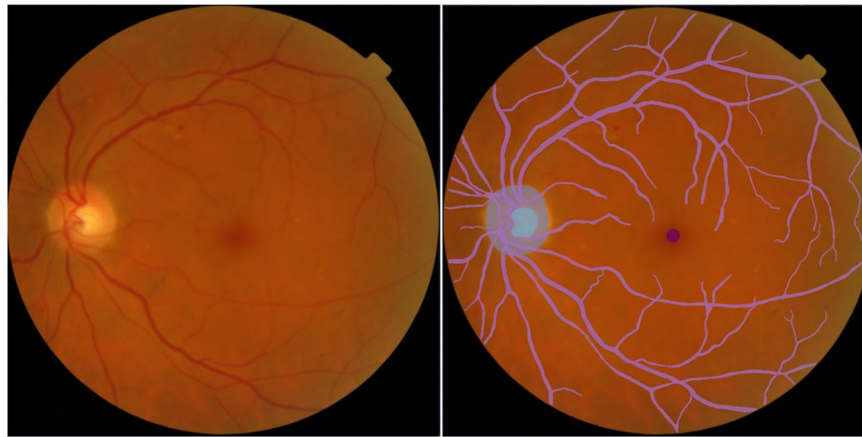


Fig. 5 Anatomical structures from image 20051205_35354_0400_PP. (Dark blue: optic disc, light blue: optic cup, dark purple: macula, light purple: vessels).



(a) No pre-processing

(b) CLAHE

(c) Contrast Maximization

(d) CLAHE + Contrast

Fig. 6 Fundus image preprocessed with the visual enhancement algorithms available in the annotation platform.

Then, two image enhancement algorithms were applied in order to assist the annotators. The first corrected illumination variations while preserving image hues, by subtracting the median-filtered image and using contrast-limited adaptive histogram equalization (CLAHE) on the brightness channel only. The second maximized the contrast by independently normalizing the intensities of each RGB channel so that they spread over their full range of values (see Fig. 6). These two preprocessing algorithms could be activated individually or in combination on the annotation interface. The decision of how and when to use these image enhancements was left to the clinician's discretion.

Reviewing and correcting preannotated segmentation maps. Labeling anatomical and pathological retinal structures at a pixel level is an intrinsically tedious task. So instead of labeling from scratch, retinologists were assigned the less laborious task of reviewing AI-generated segmentations maps (we will refer to them as *preannotated maps*) and correcting any segmentation errors. Preannotated maps were only provided for exudates, microaneurysms, hemorrhages, and blood vessels. Labeling the other anatomical biomarkers (macula, optic disc, and optic cup) was more straightforward and didn't require preannotation. As for the other pathological biomarkers (neovessels, drusen and CWS), the databases published at the time were not sufficient to train segmentation models and therefore those biomarkers were labeled from scratch.

In practice, generating the preannotated maps was entrusted to two models: one responsible for segmenting lesions³⁴, the other for segmenting vessels³⁵. Both models were state-of-the-art in 2018. The final segmentation thresholds of both models were slightly reduced to favor false positives over false negatives. Indeed, we anticipated that reviewing all preannotated structures and removing the incorrect ones would be an easier task than spotting and annotating all those missing. Conversely, no lesions were pre-annotated for healthy images (according to MESSIDOR's DR labels), to avoid the need of removing them.

Upon reaching 100 reviewed and corrected vessel segmentations, the weights of the pre-annotation model were fine-tuned on those new labels. The resulting improvements to the automated segmentation maps further reduced by 10% the average time required to annotate vessels.

Instructions provided to retinologists. In preparation of the annotation process, we separated the 10 biomarkers into 4 categories: **1.** red lesions (micro-aneurysms, hemorrhages, and neovessels), **2.** bright lesions (exudates,

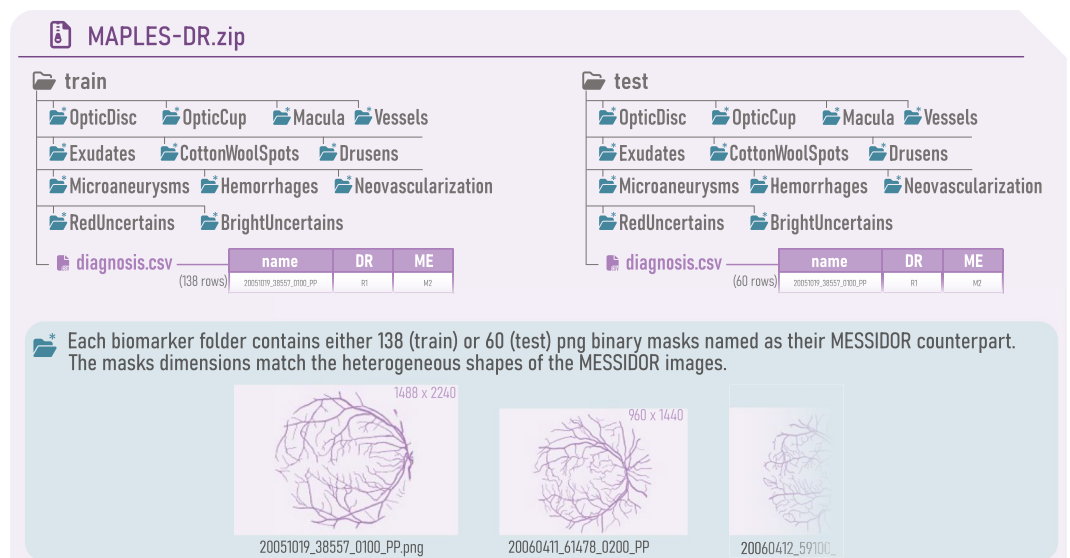


Fig. 7 Overview of the MAPLES-DR principal archive: MAPLES-DR.zip.

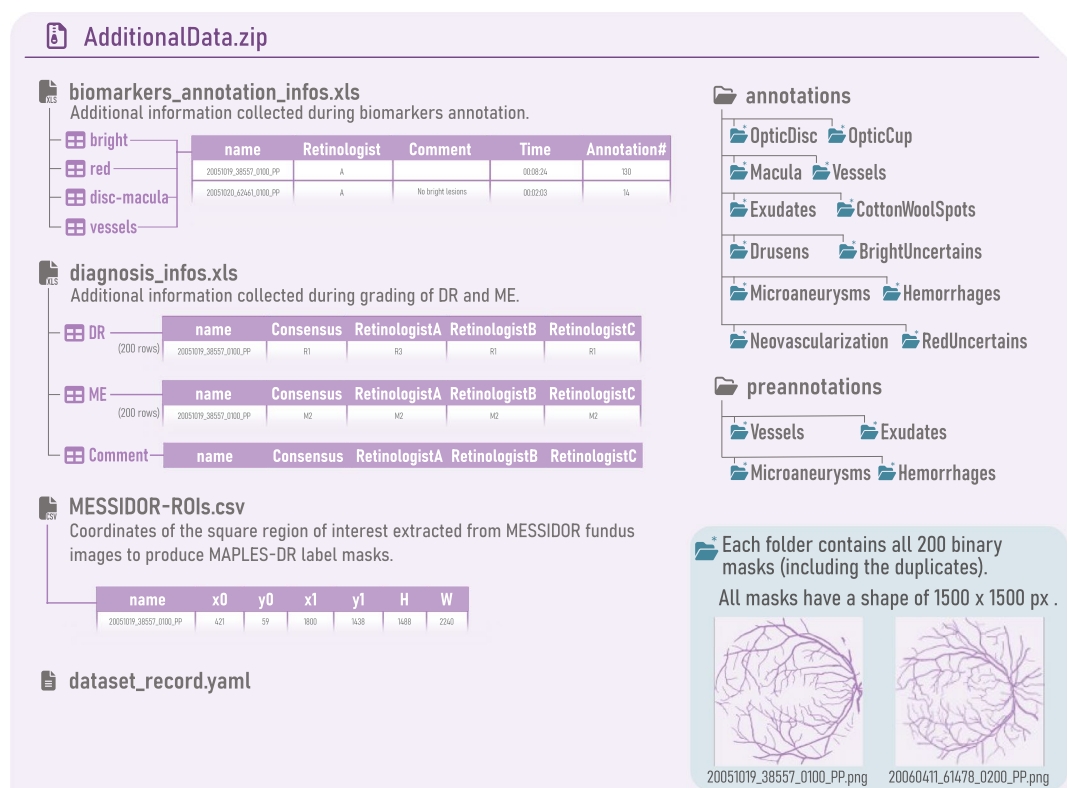


Fig. 8 Overview of MAPLES-DR complementary archive: AdditionalData.zip.

drusen, cotton wool spots), 3. vessels, and 4. other anatomical biomarkers outside vessels (optic disc, optic cup, macula). An additional “uncertain” channel was added to the two pathological biomarker categories (i.e. *uncertain bright* and *uncertain red*) in anticipation of ambiguous cases. Here, annotators could annotate any structures that appeared pathological, but whose exact nature was unclear.

Each retinologist was assigned one of four categories and focused on annotating only the corresponding biomarkers. Consequently, each of the 198 images was seen by several retinologists but each type of biomarker was only reviewed by a single expert: one segmenting vessels, another the red lesions, etc. By splitting the annotation tasks and specializing each clinician in the labeling of a few biomarkers, we intended to simplify the learning curve for the annotation tools and speed up the entire annotation process. This measure, combined

with preannotated labeling, reduced the cumulative time needed to annotate an image to an average of 22 minutes.

Besides the list of biomarkers to label and a list of recommendations on using the annotation tools, no explicit labeling instructions were provided to clinicians. We trusted their diagnostic expertise and did not provide instructions on the clinical definition of biomarkers or the level of detail expected. Yet, it is worth noting that the platform's design induced implicit guidelines. Lesions in the preannotated maps were small (the median lesion diameter was 13 pixels) and numerous; this influenced the annotators to label each lesion individually (instead of circling the general area) and to pay attention to all lesions, even the smallest ones. As Radsch *et al.*³⁶ recently reported, providing expert annotators with exemplary images instead of extended text descriptions significantly improves the quality of biomedical annotations.

Data Records

The MAPLES-DR dataset is available as a figshare dataset³⁷ and contains two archives: [MAPLES-DR.zip](#) and [AdditionalData.zip](#). The fundus images themselves are not included in these archives, as requested by the MESSIDOR consortium, but they are freely available to researchers who request them on the [Consortium's website](#).

MAPLES-DR.zip. The first archive provides the aggregated and final data of MAPLES-DR. It is intended to be used in direct conjunction with the [MESSIDOR-1 archive](#) to provide DR and ME labels and segmentation maps to use with their fundus images.

The archive `MAPLES-DR.zip` contains two folders: `train/` and `test/`, which are organized with a common structure (see Fig. 7 for an overview). Each includes a csv table named `diagnosis.csv` which associates the image names (following the MESSIDOR naming convention) with their corresponding DR and ME consensus grades. Each folder contains 12 subfolders: one for each segmented biomarker: the four anatomical biomarkers, the six pathological biomarkers, as well as `RedUncertains` and `BrightUncertains` for the ambiguous pathological structures. In those subfolders, the segmentation maps are saved as binary images in PNG format with the same naming convention and at the same resolution as their MESSIDOR counterparts. Note that because the MESSIDOR images vary in dimensions (ranging from 960×1440 up to 1536×2304 pixels), the images in `MAPLES-DR.zip` do as well.

To divide the dataset into train and test sets, a multilabel vector was built for each image indicating the presence or absence of each type of lesion. The sets were then obtained by applying iterative multilabel-shuffle stratification as introduced by Sechidis *et al.*³⁸. The resulting training and testing sets comprise, respectively, 138 and 60 samples. However, one segmentation of the macula is missing from the training set because the fundus image is centered on the disk, and four training and two testing maps of the optic cup are also unavailable as their boundaries were too fuzzy to be segmented. The corresponding files were simply removed from the folders: `train/Macula`, `train/OpticCup` and `test/OpticCup`, reducing their file counts to 137, 134, and 58 respectively.

AdditionalData.zip. The second archive contains all additional information and data collected throughout the annotation process, including the two image duplicates (see Fig. 8 for an overview). It is a complementary resource to the main archive, intended for multiple applications. For instance, these data can serve to conduct variability studies on non-aggregated DR and ME diagnoses or on the segmentation of retinal structures in the duplicate images; to better appreciate the annotation work by consulting the preannotated maps or the time spent correcting them; or to complement the annotations themselves with the textual comments left by retinologists.

It contains the following files and directories:

- `biomarkers_annotations_infos.xls`: identify which Retinologist performed the annotation of a given biomarker category, the Time spent on each annotation (in seconds), any Comment they left, and the Annotation# rank (1 for the first image annotated, 200 for the last).
- `diagnosis_infos.xls`: contains the DR and ME grades annotated by each Retinologist and the consensus they reached. It also includes any Comment left by the retinologists while grading.
- `pre_annotations/`: contains the automatic segmentation of Vessels, Exudates, Hemorrhages, and Microaneurysms provided as preannotations to the retinologists.
- `annotations/`: stores 12 subfolders (one for each biomarker), which in turn contains all 200 segmentation maps annotated by the retinologists, without distinction between the train, test and duplicate sets. The images are stored as PNG binary masks at the resolution at which they were annotated (1500×1500 pixels) using the ROIs provided in the `MESSIDOR-ROIs.csv`.
- `MESSIDOR-ROIs.csv`: provides the bounding boxes extracted from the MESSIDOR images to obtain square regions of interest and remove as much of the blank borders as possible around the circular fundus area. For each image identified by a name, its bounding box is stored as a top-left (x_0, y_0) and bottom-right (x_1, y_1) pair of coordinates in pixels. The original resolution (W, H) of the MESSIDOR fundus images is also included (resp. width and height in pixel). Those coordinates are necessary in order to either pad and resize the `annotations/` and `pre_annotations/` maps to match MESSIDOR resolution, or conversely, to crop and resize the MESSIDOR images to match the 1500×1500 resolution of MAPLES-DR segmentation maps.
- `dataset_record.yaml`: lists the biomarkers names, the image filenames in the train, test and duplicates sets, as well as the filename of the samples that are missing an optic cup or macula segmentation map (resp. `no_cup` and `no_macula`).

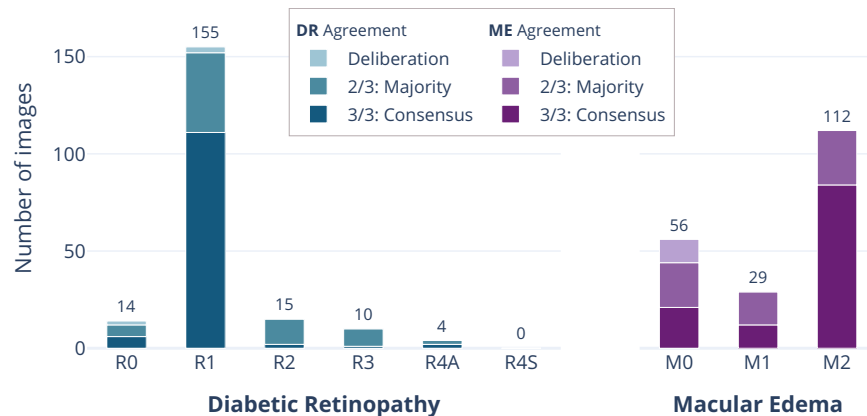


Fig. 9 Image counts for each DR and ME grade in MAPLES-DR and number of retinologists agreeing with the final grade. (Deliberation was required when all three retinologists were in disagreement).

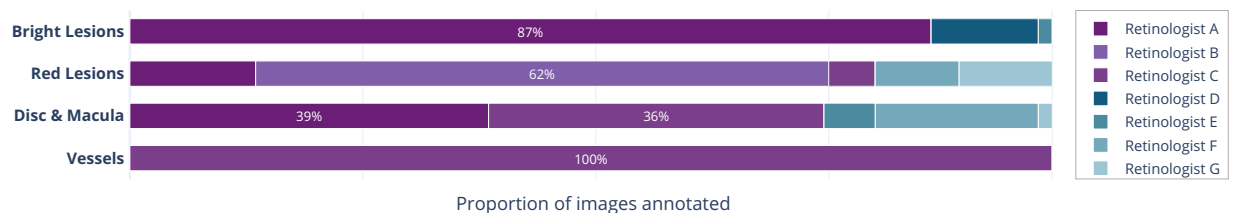


Fig. 10 Proportion of images annotated by each retinologist per category of biomarker.

Technical Validation

DR and ME grades: overview. Of the 198 images selected from MESSIDOR, all were judged to be of sufficient quality to receive a DR grade. Only one was deemed insufficient for ME grading. Our team of retinologists diagnosed a large majority of images as stage *R1: Mild DR* and none as stage *R4S: Stable Treated Proliferative* (cf. Fig. 9). From this perspective, the discrepancy between the grade distribution published in MESSIDOR that we used to select the 200 images (cf. Fig. 2) and the MAPLES-DR grades may seem significant, but two factors must be considered to understand this disparity. First, the visualization tools available in our annotation platform promote high sensitivity of lesions detection: MAPLES-DR retinologists reported that CLAHE preprocessing was particularly helpful for spotting small red lesions (i.e. microaneurysms and small hemorrhages) at first glance. Thanks to these tools, they identified enough microaneurysms in half of the MESSIDOR R0 (healthy) images to reclassify them as R1. Second, to be diagnosed as R2 in MAPLES-DR, an image must contain at least 4 hemorrhages (according to international grading guidelines), but it only needs to contain 1 hemorrhage or 5 microaneurysms to receive the same grade in MESSIDOR. Due to this difference between the two grading systems, many images classified as R2 or R3 in MESSIDOR were reclassified as R1 in MAPLES-DR. The opposite trend is seen for the macular edema scores: Although most of the selected images were classified as M0 in MESSIDOR, many were reclassified as M2 in MAPLES-DR because the former grading focuses on exudates, while the latter grading also considers microaneurysms and hemorrhages within the macula. These results underline the impact of the grading systems on global pathological labels and validate *a posteriori* our choice to reannotate the images in a system comparable to the international standard.

Our DR and ME grading procedure required a deliberation phase when all three retinologists were in disagreement. This step was necessary in five cases for DR and 12 cases for ME, and the deliberation always resulted in a low-severity grade: R0, R1 or M0 (see Fig. 9). Although not negligible, the disagreement rate between our three retinologists was consistent with the scores found in the literature: the quadratic Cohen kappa scores between individual and consensus grades for our three retinologists were 0.873, 0.862, and 0.619, whereas Krause *et al.*³¹ measured an average of 0.87 for the same metric. More recently Teoh *et al.*³² measured an intra-group Gwet AC2 coefficient of 0.628, while our retinologists achieved 0.964 (CI 95%: 0.947–0.981) for DR and 0.844 (CI 95%: 0.799–0.888) for ME.

Retinal structures: overview of annotation work. The pixel-wise annotation of DR biomarkers was performed primarily by the three retinologists who also graded DR and ME: one specialist annotated the bright lesions on 87% of the images, the second labeled the red lesions on 62% of them, while the third performed the segmentation of the vessels for all 198 images. The other four retinologists contributed mainly to the annotation of the optic disc and macula, as well as a few red and bright lesions (cf. Fig. 10). In total, the team dedicated 69 hours to reviewing preannotated structures, erasing false positives, manually adding annotations that were missing from the preannotated maps, and correcting lesions that were correctly segmented but misclassified (e.g., microaneurysms instead of hemorrhages).

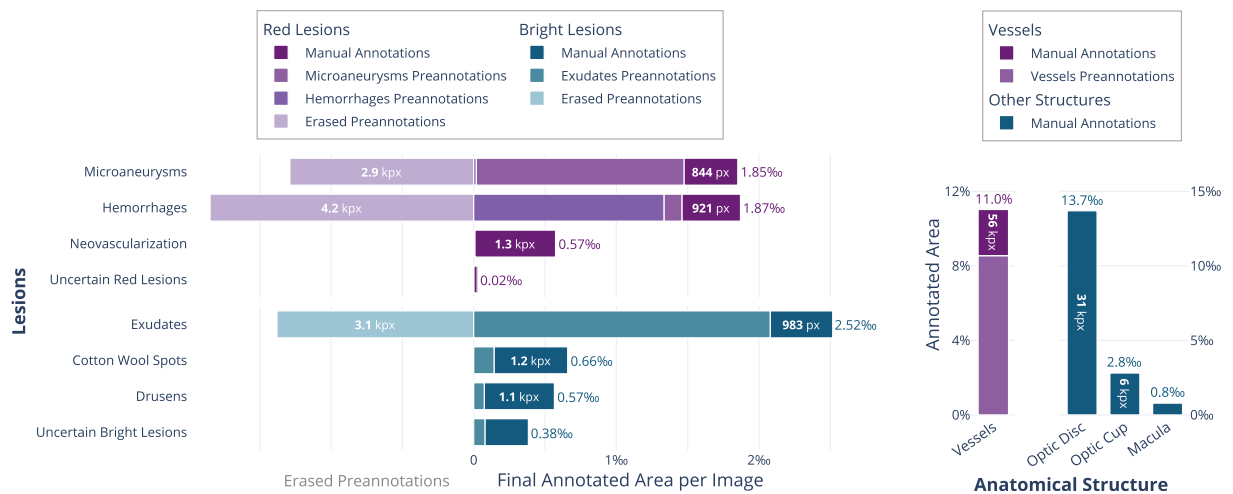


Fig. 11 Overview of the annotation work performed by the retinologists for each category of biomarkers: red lesions, bright lesions, vessels, and other anatomical structures. Manually added annotations appear darker while manually erased pre-annotations appear lighter. Both are labeled with the average number of corrected pixels per image. The final area covered by each type of biomarker after correction is displayed as a fraction of the total image area of the dataset.

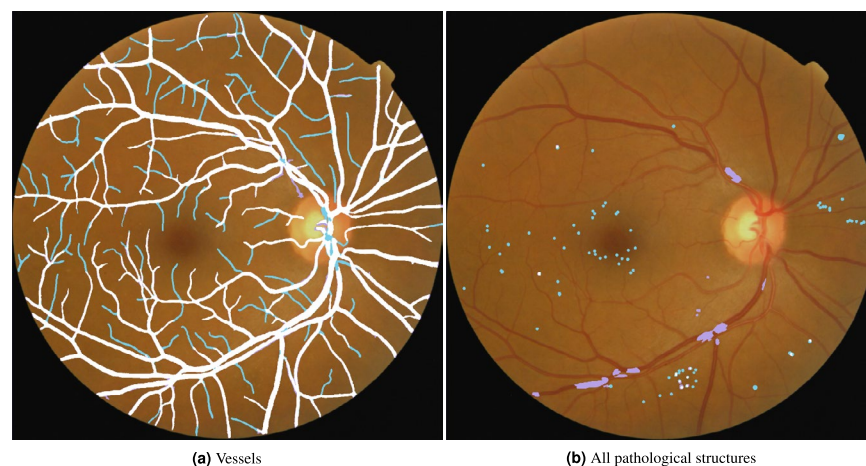


Fig. 12 Examples of corrections performed on image 20060412_61593_0200_PP when reviewing its preannotated segmentation maps. (White: unchanged preannotation, teal: manually added segmentation, purple: manually removed preannotation).

The macula and optic disc (including the cup) were the fastest to be annotated, with an average of 2 minutes per image, followed by vessel segmentation, with an average of 6 minutes per image. For the latter, preannotation appears to have been a valuable aid, since 77.5% of the pixels found in the vascular maps published in MAPLES-DR were preannotated and less than 8.3% of pixels from the initial preannotation maps were manually erased as false positive. However, the vessel labeling process still required substantial manual corrections: On average, an area of 56k pixels was manually added to each segmentation map (cf. Fig. 11), most of which consisted of small vessels (see Fig. 12a for an example of manual corrections of the vessels preannotations). In terms of image area, anatomical structures account for most of the MAPLES-DR annotations because, in addition to being typically larger than lesions, they are present in all images. Note that there are six images in which the optic cup segmentation is missing because its boundaries were too uncertain, and one image in which the macula is not visible as the image is centered on the optic disc.

Despite covering only a small fraction of the MAPLES-DR images (0.43% of pixels for red lesions and 0.41% for bright lesions), pathological structures required the greatest annotation effort. The time spent on each image varied greatly: from a few dozen seconds for healthy images to an hour for the most severe cases. Similarly to vessels, manual annotation of pre-segmented lesions (microaneurysms, hemorrhages, and exudates) relied heavily on the automatic segmentations. Only a quarter of the pixels were manually added, while the rest came directly from the preannotated maps. An analysis of the size distribution of lesions added manually by retinologists reveals that many small lesions (i.e. microaneurysms, exudates, or drusens) were annotated as 10px diameter circles. This shape corresponds to the default configuration of the annotation tool and is not representative of

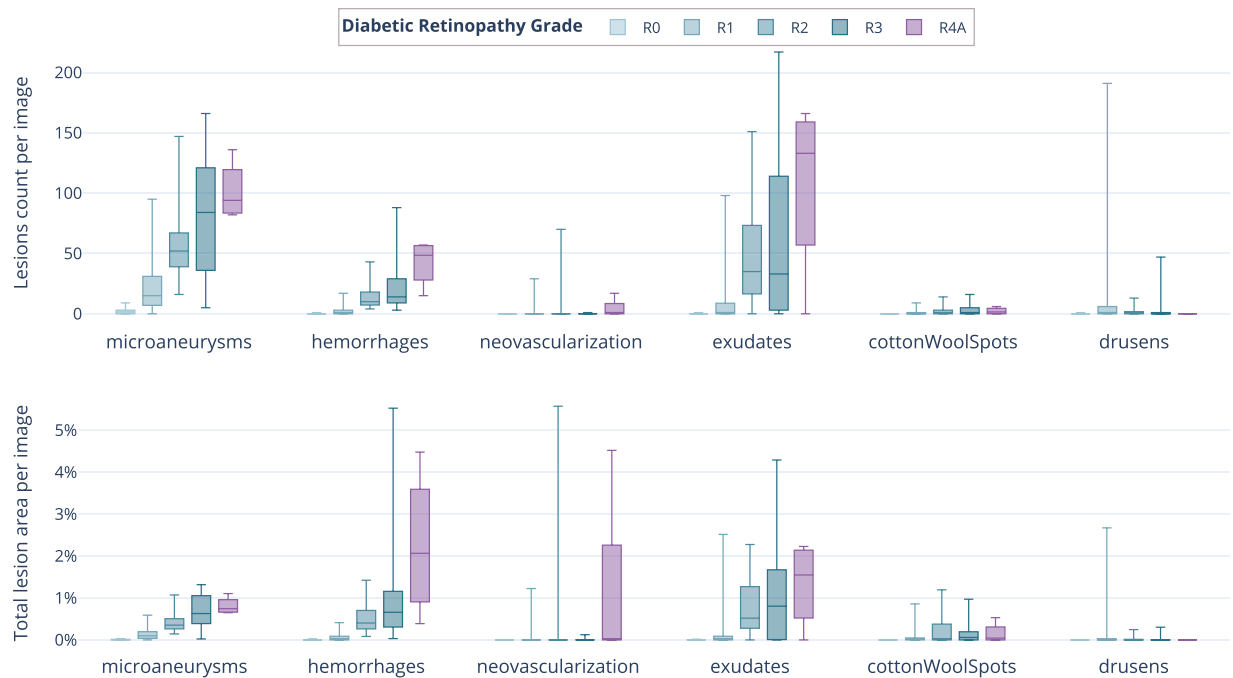


Fig. 13 Distribution of the number of individual lesions and their total area per image for each type of pathological structure, grouped by DR severity.

the exact shape of these small lesions. Unlike vessels, preannotation lesion maps contained more false positives and required a more careful review by retinologists to eliminate them. More than 58% of the preannotated hemorrhages were erased, as were 47% of the microaneurysms and 40% of the exudates (cf. Fig. 11). Most of the erased hemorrhages and microaneurysms were vessels erroneously detected as pathological, as seen in Fig. 12b. Overall, the annotation of pathological structures took an average of 6 minutes for bright lesions and 10 minutes for red lesions.

Retinal Lesions: correlation with DR grade. To assess the relevance of the annotations of pathological structures, we calculated the number of individual lesions per image, as well as their total area, and compared the distribution of both with respect to the severity of diabetic retinopathy (cf. Fig. 13). The general trend of these distributions matches the clinical intuition: the number of lesions per image and the area they cover increase with the severity of the pathology. Furthermore, the distribution of individual lesions is in line with the clinical definition of each grade. The number of microaneurysms, characteristic of low-severity grades, increases substantially between R0 and R1, and again between R1 and R2. For hemorrhages and exudates, whose count distinguishes the more severe stages, the most significant transitions are between stages R1 and R2, and between stages R3 and R4A. Similarly, the average surface area covered by the neovessels increases between stages R3 and R4A. Such transitions are absent in the cotton wool spots and drusen distributions, as there are too few of them in MAPLES-DR to observe any trends ($<0.01\%$ of total pixels).

Retinal structures: semantic segmentation baseline. The learnability of MAPLES-DR labels of retinal structures was tested on a semantic segmentation task by training a simple UNet model to jointly segment them all as a multiclass map. We used a straightforward training protocol: the model was trained with a learning rate of 0.003, using the Dice coefficient as the loss function and stochastic gradient descent (SGD) as the optimizer. To accelerate the training, the encoder's initial weights were pretrained on ImageNet.

The semantic segmentation performances of the model trained on the MAPLES-DR training set and tested on its test set are summarized in Fig. 14. The model was able to successfully segment the four anatomical structures, as well as exudates, microaneurysms, and hemorrhages. Segmentation performances for these last two classes may appear low but are, in fact, comparable to the scores obtained when learning to segment these lesions with other public databases. Actually, the average segmentation precision of microaneurysms of a model trained and tested on MAPLES-DR – where microaneurysms are particularly well represented – is significantly higher (56%, see Fig. 14(a)) than that obtained with the model that won the IDRiD challenge (50.2%, see Table 2). Unfortunately, not all lesions provided in MAPLES-DR are as easily learnable. Our simple model was unable to provide satisfactory segmentation for CWS, drusen, and neovessels. Indeed, because MAPLES-DR contains few images of severe or proliferative DR, examples of these lesions are rare in its training set and even rarer in its test set.

Retinal lesions segmentation: training on MAPLES-DR, testing on public datasets. To put MAPLES-DR's relevance to model learning in perspective with other popular public datasets, we trained five

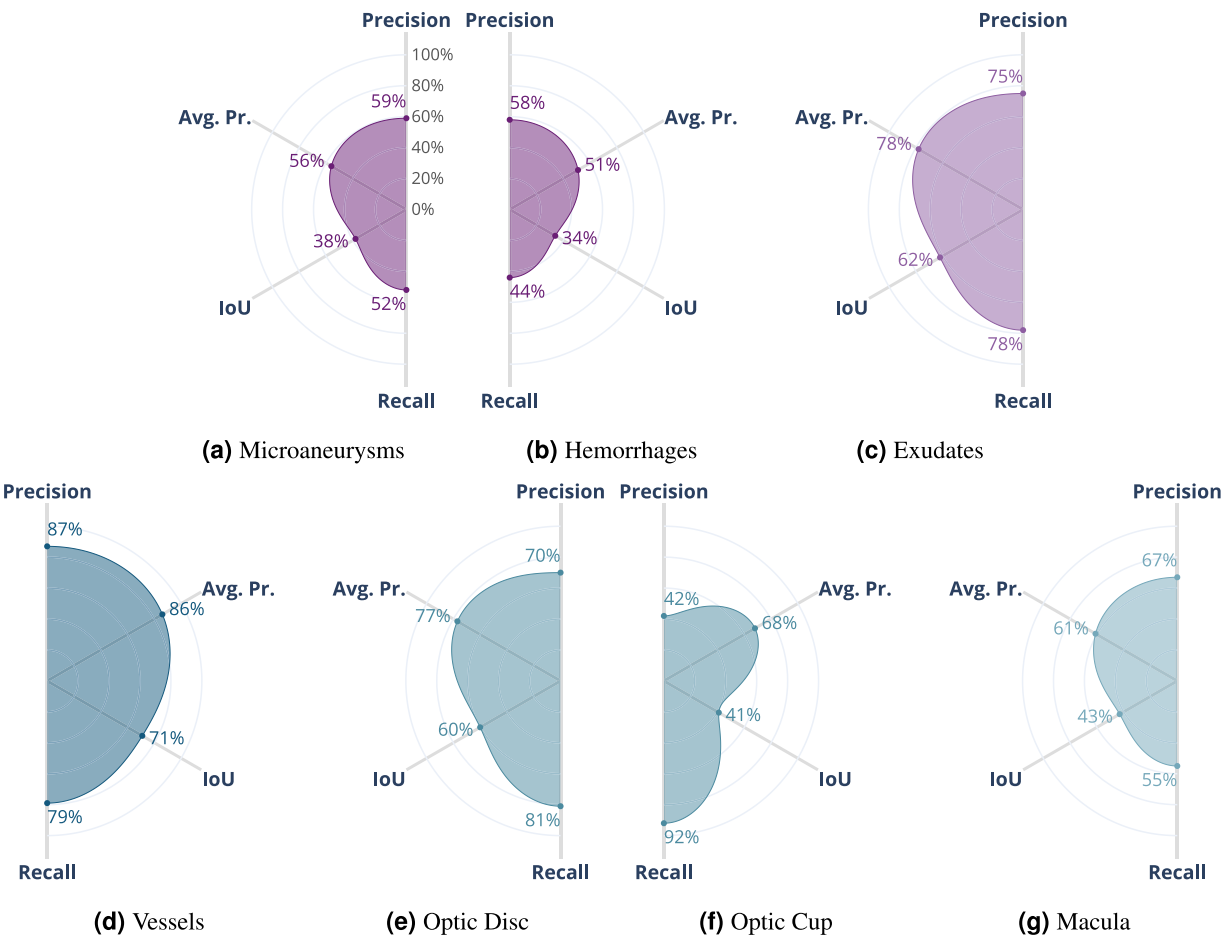


Fig. 14 Summary of the semantic segmentation performances of a simple UNet model trained to segment all retinal structures available in MAPLES-DR.

Training set		Performance by test set (mIoU)					Avg. Rank
name	size	IDRiD	MAPLES-DR	DDR	FGADR	Ret. Les.	
IDRiD	54	57.6%	37.5%	35.6%	28.4%	26.0%	2.2
MAPLES-DR	118	39.7%	46.8%	31.0%	26.9%	27.8%	2.8
DDR	450	48.6%	35.4%	43.0%	26.4%	24.8%	3.2
FGADR	1096	33.0%	28.8%	31.7%	47.4%	25.5%	3.4
Retinal Lesions	948	28.6%	27.3%	28.6%	28.4%	48.2%	3.4

Table 1. Performance comparison of lesions segmentation models trained on MAPLES-DR and four other public datasets of fundus lesions. Performance metric is mIoU. The training datasets are sorted by their average ranking on all test sets.

models to segment retinal lesions (i.e. Micro-Aneurysms, Hemorrhages, Exudates and CWS) on MAPLES-DR, IDRiD²⁵, DDR²⁶, FGADR²³, and Retinal Lesions²⁴. All models were then evaluated on the individual test sets associated with each dataset. The measured mIoUs for every combination of train and test sets are compiled in Table 1. For each test set, the model that achieves the best performance is the one trained on the corresponding train set. This result was expected given the discrepancies among the characteristics of the databases with respect to fundus images (ethnicity, image quality and resolution, colorimetry, etc) and labeling style (coarse segmentations for Retinal Lesions, precise ones for MAPLES-DR, IDRiD, and DDR and a mixture of them for FGADR)³⁹. In spite of those differences, models trained on one dataset can, to varying degrees, generalize to the others. We measured this ability by ranking the models by mIoU on each test set and then averaging those ranks. The model trained on MAPLES-DR achieves the second-best average rank of 2.8, behind IDRiD, whose average rank is 2.2. Interestingly, those were the two models trained with the smallest numbers of training samples, but on datasets containing only good-quality images and labeled with precise segmentations of lesions.

Model	MA	HEM	EX	CWS	Mean
iFLYTEK-MIG (Winner of IDRiD challenge)	0.502	0.559	0.874	0.659	0.649
UNet trained on IDRiD	0.433	0.616	0.818	0.596	0.616
UNet pretrained on MAPLES-DR, finetuned on IDRiD	0.432	0.685	0.816	0.661	0.649

Table 2. Effect of MAPLES-DR pretraining on the average precision score of lesion segmentation on the IDRiD test set. The last column present the mean over all lesions of the average precision scores.

Retinal lesions segmentation: pretraining with MAPLES-DR. As a final validation step, we illustrate how MAPLES-DR can be used in conjunction with another existing public dataset to improve segmentation performance. Because of the aforementioned distribution misalignments between datasets, it is often not relevant to simply combine training sets, and transfer learning is generally a better choice to take advantage of features learned on one dataset to improve training on another. We tested the benefit of MAPLES-DR in this context by training two simple UNet models on the IDRiD retinal lesions dataset. While both models were initialized with weights pre-trained on ImageNet, one was also pretrained on the MAPLES-DR training set before being fine-tuned on IDRiD. In both cases, we used the same basic training protocol as in the previous section. The resulting average precision scores for the segmentation of microaneurysms, hemorrhages, exudates, and cotton wool spots are presented in Table 2, along with the best model from the IDRiD competition leaderboard.

While pretraining with MAPLES-DR did not have much effect on microaneurysms and exudates, it significantly improved the segmentation of hemorrhages and cotton wool spots by a margin of 11%. For these two lesions, the model pre-trained on MAPLES-DR even outperformed the winning submission of the IDRiD challenge. Such performances are remarkable considering the rudimentary training method we used, compared to the arsenal of performance optimization techniques usually involved in challenge submissions: k-fold cross-validation, model ensembles, sophisticated regularizations, extensive hyperparameter tuning, etc. All these techniques would very likely significantly improve the performances we obtained, but are outside the scope of this study. Still, even without them, a simple UNet pre-trained on MAPLES-DR and finetuned on IDRiD equals the IDRiD challenge winner according to the competition ranking metric (i.e. the mean over all four lesions of the average precision score).

Usage Notes

The fundus images corresponding to the diagnostic labels and biomarker segmentation maps of MAPLES-DR are the property of the MESSIDOR consortium and are freely available from [the Consortium's website](#). Users must be careful to download the [MESSIDOR original](#) dataset, not [MESSIDOR-2](#) (MAPLES-DR uses several images that were removed in MESSIDOR-2). For more information on the download procedure, refer to the [dedicated page of the MAPLES-DR documentation](#).

The pathological segmentation structures present in MAPLES-DR are unevenly represented: while microaneurysms, hemorrhages and exudates are numerous, this is not the case for CWS, drusens, and neovessels (see Fig. 13). For the latter biomarkers, MAPLES-DR might be more appropriate as a pre-training set (cf. Section pre-training with MAPLES-DR) rather than as a training or test set. In addition, to identify neovascularization examples in MAPLES-DR images, one can rely on a DR grade of R4A or on the annotating clinician's comments pertaining to vessels.

We also wish to draw the reader's attention to the non-negligible number of bright lesions annotated as `BrightUncertain` when they were clearly pathological but difficult to classify as drusen, exudates or CWS. We recommend including them in the training and testing sets of models that segment or detect bright lesions without classifying them. For image-specific information on bright lesions annotated as uncertain, please consult the comments left by the annotators.

Finally, the MAPLES-DR segmentation maps were padded and resized to heterogeneous sizes and resolutions to match the original MESSIDOR fundus images. However, libraries specialized in training segmentation models commonly expect a standardized image size. We advise researchers who wish to use MAPLES-DR in such a context to crop and resize MESSIDOR fundus images with the same ROI we used to annotate them. We realize that such operations require a certain amount of boiler plate code; therefore, we have published a Python library, `maples_dr`, which automates downloading the MAPLES-DR labels, and extracting and preprocessing the MESSIDOR fundus images to a uniform image resolution. See the Code Availability section for more information.

Code availability

MAPLES-DR annotations must be paired with MESSIDOR fundus images to be exploited to their full potential. To simplify the joint usage of both datasets, we developed a python package named `maples_dr`, which is available on [pypi](#) and [GitHub](#). The key features of this library are: loading and exporting the fundus images with their annotations in formats suited for machine learning applications, preprocessing the fundus images, and providing a pythonic access to all MAPLES-DR additional data (retinologist comments, annotation times, preannotations, etc). Quick-start and code examples are available in the [maples_dr documentation](#), along with an API reference. The repository also contains the code that produced the figures in this paper.

The algorithms used to generate the preannotation maps cannot be made public as they were implemented with the Theano library, which is no longer maintained. For transparency purposes, the original preannotations are available in the `AdditionalData.zip` archive of MAPLES-DR. However, we have released up-to-date Python

segmentation models (trained with MAPLES-DR and other public datasets) in two Github repositories: [gabriel-lepetitaimon/fundus-vessels-toolkit](#) provides automatic segmentation and graph extraction of the retinal vasculature; [ClementPla/fundus-lesions-toolkit](#) provides automatic segmentation and visualization of retinal lesions.

Finally the annotation platform code and documentation are freely available to any research team that wishes to use it in the Github repository: [LIV4D/AnnotationPlatform](#).

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References

1. Lanzetta, P. *et al.* Fundamental principles of an effective diabetic retinopathy screening program. *Acta Diabetologica* **57**, 785–798, <https://doi.org/10.1007/s00592-020-01506-8> (2020).
2. Cheng, A. Y. Y. Canadian Diabetes Association 2013 clinical practice guidelines for the prevention and management of diabetes in Canada. *Canadian Journal of Diabetes* **37**, S1–S3 (2013).
3. Hooper, P. *et al.* Excerpt from the Canadian Ophthalmological Society evidence-based clinical practice guidelines for the management of diabetic retinopathy. *Canadian Journal of Ophthalmology* **52**, S45–S74, <https://doi.org/10.1016/j.jcjo.2017.09.027> (2017).
4. Egunsola, O. *et al.* Diabetic retinopathy screening: A systematic review of qualitative literature. *Canadian Journal of Diabetes* **45**, 725–733.e12, <https://doi.org/10.1016/j.jcjd.2021.01.014> (2021).
5. Avidor, D., Loewenstein, A., Waisbourd, M. & Nutman, A. Cost-effectiveness of diabetic retinopathy screening programs using telemedicine: a systematic review. *Cost Effectiveness and Resource Allocation* **18**, <https://doi.org/10.1186/s12962-020-00211-1> (2020).
6. Raman, R. *et al.* Fundus photograph-based deep learning algorithms in detecting diabetic retinopathy. *Eye* **33**, 97–109, <https://doi.org/10.1038/s41433-018-0269-y> (2019).
7. Dugas, E., Jorge, J. & Cukierski, W. Diabetic retinopathy detection. <https://kaggle.com/competitions/diabetic-retinopathy-detection> (2015).
8. Decencière, E. *et al.* Feedback on a publicly distributed image database: the MESSIDOR database. *Image Analysis & Stereology* **33**, 231–234, <https://doi.org/10.5566/ias.1155> (2014).
9. Gulshan, V. *et al.* Performance of a deep-learning algorithm vs manual grading for detecting diabetic retinopathy in India. *JAMA Ophthalmology* **137**, 987, <https://doi.org/10.1001/jamaophthalmol.2019.2004> (2019).
10. Phillips, P. J., Hahn, C. A., Fontana, P. C., Broniatowski, D. A. & Przybocki, M. A. Four principles of explainable artificial intelligence. Tech. Rep. NISTIR 8312, National Institute of Standards and Technology, Gaithersburg, Maryland <https://doi.org/10.6028/NIST.IR.8312> (2020).
11. Wilkinson, C. P. *et al.* Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology* **110**, 1677–1682, [https://doi.org/10.1016/S0161-6420\(03\)00475-5](https://doi.org/10.1016/S0161-6420(03)00475-5) (2003).
12. Zachariah, S., Wykes, W. & Yorston, D. Grading diabetic retinopathy (DR) using the Scottish grading protocol. *Community Eye Health* **28**, 72 (2015).
13. Boucher, M. *et al.* Evidence-based Canadian guidelines for tele-retina screening for diabetic retinopathy: recommendations from the Canadian Retina Research Network (CR2N) Tele-Retina Steering Committee. *Canadian Journal of Ophthalmology* **55**, 14–24, <https://doi.org/10.1016/j.jcjo.2020.01.001> (2020).
14. Staal, J., Abramoff, M. D., Niemeijer, M., Viergever, M. A. & Van Ginneken, B. Ridge-based vessel segmentation in color images of the retina. *IEEE Transactions on Medical Imaging* **23**, 501–509, <https://doi.org/10.1109/TMI.2004.825627> (2004).
15. Budai, A. *et al.* Robust vessel segmentation in fundus images. *International Journal of Biomedical Imaging* **2013** <https://doi.org/10.1155/2013/154860> (2013).
16. Jin, K. *et al.* Fives: A fundus image dataset for artificial intelligence based vessel segmentation. *Scientific Data* **9**, 475, <https://doi.org/10.1038/s41597-022-01564-3> (2022).
17. Lyu, X., Cheng, L. & Zhang, S. The retina benchmark for retinal vascular tree analysis. *Scientific Data* **9**, 397, <https://doi.org/10.1038/s41597-022-01507-y> (2022).
18. Orlando, J. I. *et al.* Refuge challenge: A unified framework for evaluating automated methods for glaucoma assessment from fundus photographs. *Medical Image Analysis* **59**, 101570 (2020).
19. Kovalyk, O. *et al.* Papila: Dataset with fundus images and clinical data of both eyes of the same patient for glaucoma assessment. *Scientific Data* **9**, 291, <https://doi.org/10.1038/s41597-022-01388-1> (2022).
20. Kumar, J. H. *et al.* Chakṣu: A glaucoma specific fundus image database. *Scientific Data* **10**, 70, <https://doi.org/10.1038/s41597-023-01943-4> (2023).
21. Lin, L. *et al.* The SUSTech-SYSU dataset for automated exudate detection and diabetic retinopathy grading. *Scientific Data* **7**, <https://doi.org/10.1038/s41597-020-00755-0> (2020).
22. Decencière, E. *et al.* Teleophtha: Machine learning and image processing methods for teleophthalmology. *IRBM* **34**, 196–203, <https://doi.org/10.1016/j.irbm.2013.01.010> (2013). Special issue : ANR TECSAN : Technologies for Health and Autonomy.
23. Zhou, Y., Wang, B., Huang, L., Cui, S. & Shao, L. A benchmark for studying diabetic retinopathy: segmentation, grading, and transferability. *IEEE Transactions on Medical Imaging* **40**, 818–828, <https://doi.org/10.1109/TMI.2020.3037771> (2020).
24. Wei, Q. *et al.* Learn to segment retinal lesions and beyond. In *2020 25th International Conference on Pattern Recognition (ICPR)*, 7403–7410 <https://doi.org/10.1109/ICPR48806.2021.9412088> (IEEE, 2021).
25. Porwal, P. *et al.* Indian diabetic retinopathy image dataset (IDRiD): a database for diabetic retinopathy screening research. *Data* **3**, 25, <https://doi.org/10.3390/data3030025> (2018).
26. Li, T. *et al.* Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening. *Information Sciences* **501**, 511–522, <https://doi.org/10.1016/j.ins.2019.06.011> (2019).
27. Abramoff, M. D. *et al.* Automated Analysis of Retinal Images for Detection of Referable Diabetic Retinopathy. *JAMA Ophthalmology* **131**, 351–357, <https://doi.org/10.1001/jamaophthalmol.2013.1743> (2013). https://jamanetwork.com/journals/jamaophthalmology/articlepdf/1668203/ecs120076_351_357.pdf.
28. Ruamviboonsuk, P., Teerasuwanajak, K., Tiensuwan, M. & Yuttitham, K. Interobserver agreement in the interpretation of single-field digital fundus images for diabetic retinopathy screening. *Ophthalmology* **113**, 826–832, <https://doi.org/10.1016/j.ophtha.2005.11.021> (2006).
29. Bragge, P., Gruen, R. L., Chau, M., Forbes, A. & Taylor, H. R. Screening for presence or absence of diabetic retinopathy: a meta-analysis. *Archives of Ophthalmology* **129**, 435–444, <https://doi.org/10.1001/archophthalmol.2010.319> (2011).
30. Shi, L. *et al.* Telemedicine for detecting diabetic retinopathy: a systematic review and meta-analysis. *British Journal of Ophthalmology* **99**, 823–831, <https://doi.org/10.1136/bjophthalmol-2014-305631> (2015).
31. Krause, J. *et al.* Grader variability and the importance of reference standards for evaluating machine learning models for diabetic retinopathy. *Ophthalmology* **125**, 1264–1272, <https://doi.org/10.1016/j.ophtha.2018.01.034> (2018).
32. Teoh, C. S. *et al.* Variability in grading diabetic retinopathy using retinal photography and its comparison with an automated deep learning diabetic retinopathy screening software. *Healthcare* **11**, 1697, <https://doi.org/10.3390/healthcare11121697> (2023).

33. Sasongko, M. B. *et al.* Retinal vascular tortuosity in persons with diabetes and diabetic retinopathy. *Diabetologia* **54**, 2409–2416, <https://doi.org/10.1007/s00125-011-2200-y> (2011).
34. Playout, C., Duval, R. & Cheriet, F. A novel weakly supervised multitask architecture for retinal lesions segmentation on fundus images. *IEEE Transactions on Medical Imaging* **38**, 2434–2444, <https://doi.org/10.1109/TMI.2019.2906319> (2019).
35. Lepetit-Aimon, G., Duval, R. & Cheriet, F. Large receptive field fully convolutional network for semantic segmentation of retinal vasculature in fundus images. In *Computational Pathology and Ophthalmic Medical Image Analysis*, 201–209 (Springer, https://doi.org/10.1007/978-3-030-00949-6_24 2018).
36. Radsch, T. *et al.* Labelling instructions matter in biomedical image analysis. *Nature Machine Intelligence* **5**, 273–283, <https://doi.org/10.1038/s42256-023-00625-5> (2023).
37. Lepetit-Aimon, G., Playout, C. & Cheriet, F. MAPLES-DR: MESSIDOR Anatomical and Pathological Labels for Explainable Screening of Diabetic Retinopathy. *figshare* <https://doi.org/10.6084/m9.figshare.24328660> (2023).
38. Sechidis, K., Tsoumakas, G. & Vlahavas, I. On the stratification of multi-label data. In *Machine Learning and Knowledge Discovery in Databases*, 145–158, https://doi.org/10.1007/978-3-642-23808-6_10 (Springer, Berlin, Heidelberg, 2011).
39. Playout, C. & Cheriet, F. Cross-dataset generalization for retinal lesions segmentation. *arXiv preprint arXiv:2405.08329* <https://doi.org/10.48550/arXiv.2405.08329> (2024).
40. Cao, J., Felfeli, T., Merritt, R. & Brent, M. H. Sociodemographics associated with risk of diabetic retinopathy detected by tele-ophthalmology: 5-year results of the Toronto tele-retinal screening program. *Canadian Journal of Diabetes* **46**, 26–31, <https://doi.org/10.1016/j.cjcd.2021.05.001> (2022).

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Author contributions

M.C.B. and F.C. conceptualized and designed the study. G.L.A. and C.P. designed and supervised the development of the annotation platform. M.C.B. and R.D. and M.B. annotated the dataset. G.L.A. supervised the annotation campaigns, documented its procedure, and finally exported and analyzed the dataset. C.P. performed and analyzed the trainability experiments. G.L.A. drafted the initial manuscript. All authors reviewed and approved the final manuscript as submitted.

Competing interests

The authors declare no competing interests.

Additional information

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